AI for Anomaly Detection

Lecture 3: Unsupervised Anomaly Detection



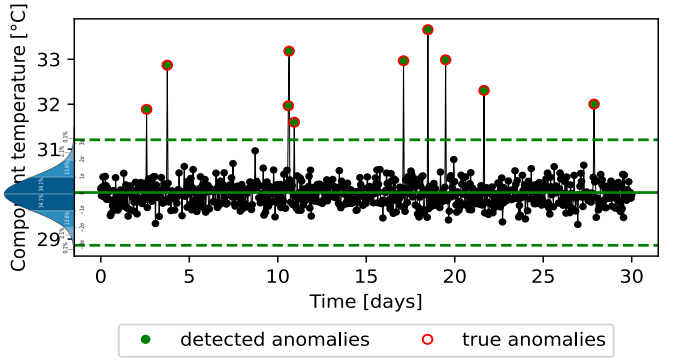


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zh aw School of Engineering

Recap

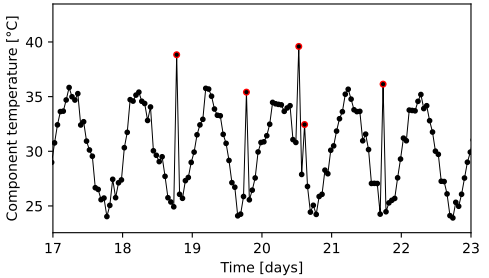
- AD in a univariate variable ➔ statistical thresholc



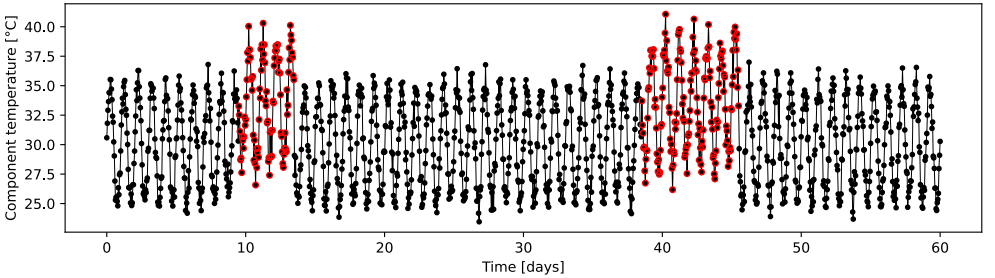
μ+3σ
μ
μ-3σ

- AD in multivariate data ➔ (linear) regression, residuals as scores.





- Supervised AD with logistic regression.
- Supervised AD with SVM.

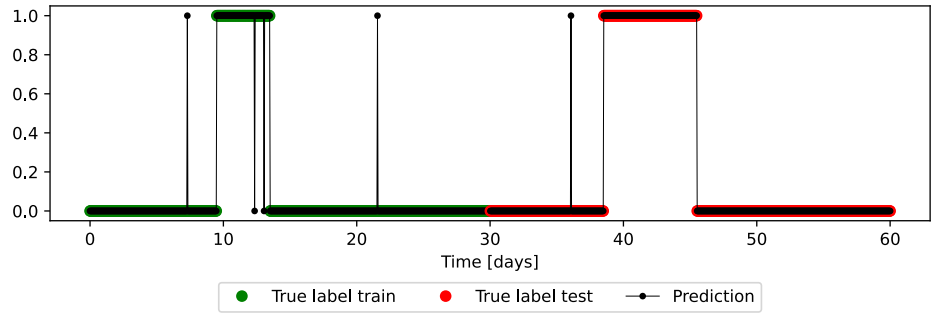


- Evaluation with precision-recall and ROC

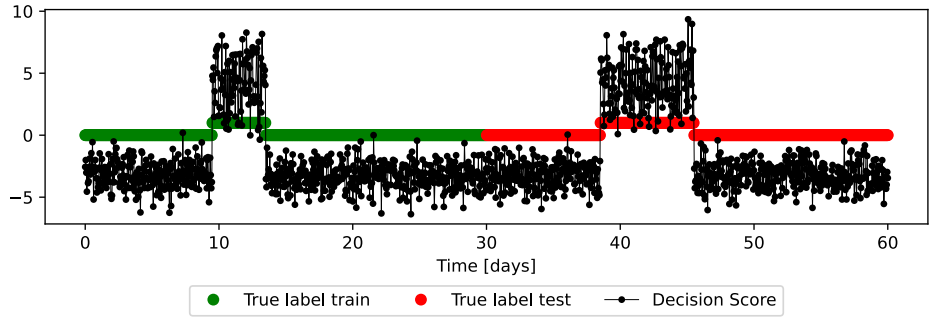
		PREDICTED	
		Positive	Negative
ACTUAL	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

Understand the algorithm

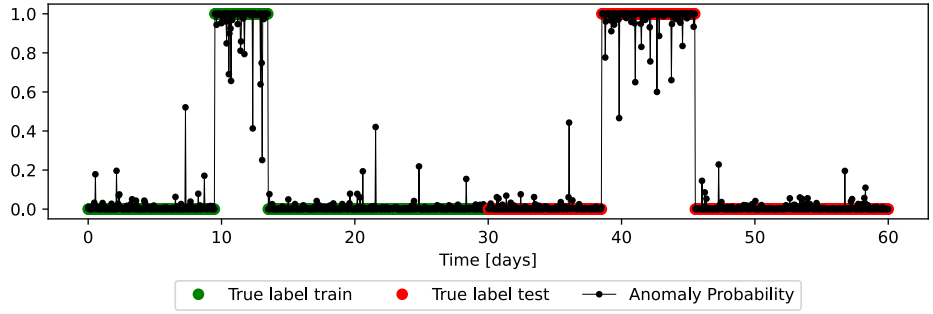
- Final classification output



- Decision score

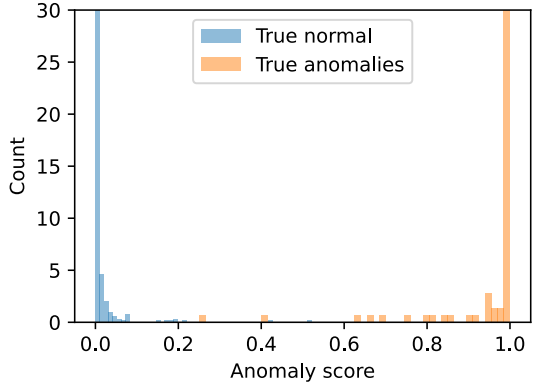


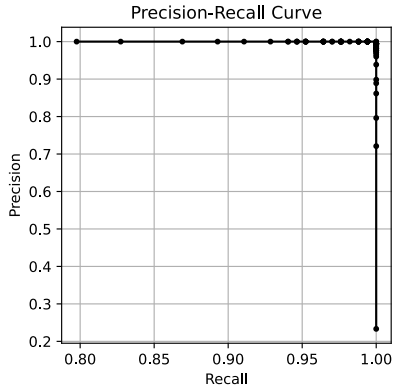
- Anomaly probability



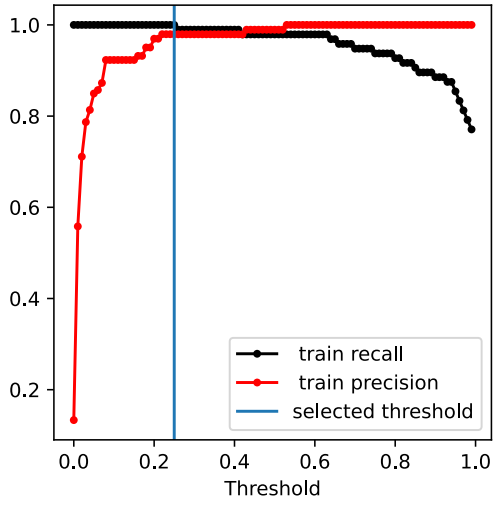
Supervised AD: use labeled data to select a threshold

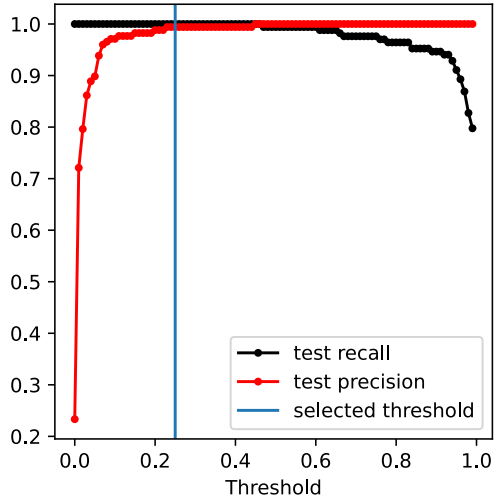
- Anomaly scores distribution





- Select threshold on training data based on use-case





Different approaches to anomaly detection

- *Supervised methods:* learn to represent normal and abnormal data based on previous examples.

  ➔ need labeled examples of normal and abnormal.

  👍 high accuracy, easier to validate, anomaly class is clear.

  👎 Disadvantage: high imbalance, high requirements from the data, novelties cannot be discovered.

- *Unsupervised methods:* detect patterns in the data that correspond to normal/ abnormal behaviour without any labels.

  ➔ no labels needed.

  👍 Low requirements on data.

  👎 Harder to achieve high performance, hard to evaluate.

- *Novelty detection (most common):* learn to represent normal data and compare unknown patterns to the representation.

  ➔ need labels only for normal examples.

  👍 normal data is usually abundant, detection independent of anomaly type.

  👎 Hard to validate, anomaly class unclear.

Understand the setting of your problem

- Many normal and abnormal examples → try **supervised** approach
  - Train-validation-test split
  - Tune the model on the validation data
  - Leave the test data only for final evaluation
- Many normal and few abnormal examples → **unsupervised** or **novelty** detection.
  - In practice you will tune and select a model on the test data
  - Real independent testing will only be possible in operation, when more test anomalies are available.
- Only normal data
  - No way for systematic model development and evaluation

➔We can **train** models without using labels, but we cannot **evaluate** them without labels.

➔Always make sure you **understand which part** of the data is used for training, validation and testing!

Today: Unsupervised methods for AD (outlier detection)

- Statistical AD (Hotteling's statistics)
- Clustering with DBSCAN
- One Class Support Vector Machine
- Exercise
- Time to ask questions about the data sets.

Unsupervised AD methods

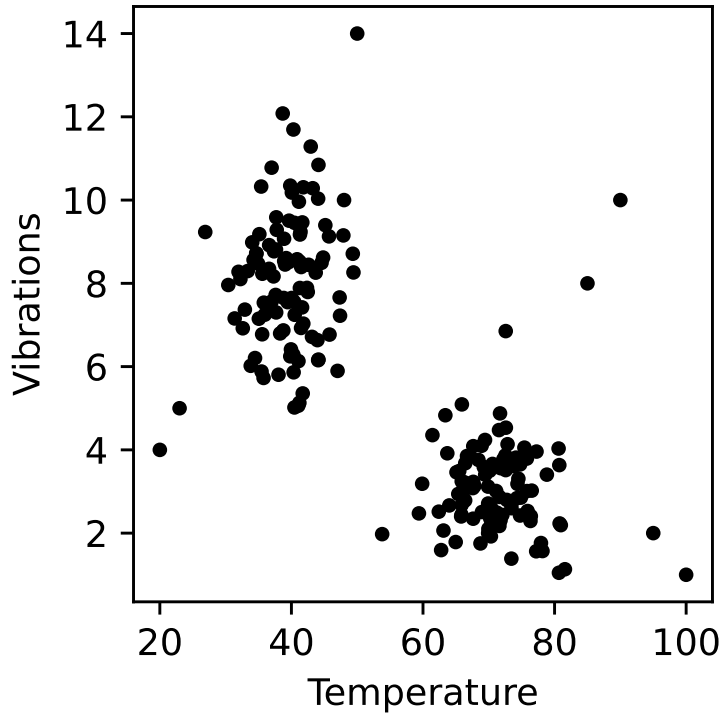
In many real-world systems we have no labels at all ➔ detect anomalies with unsupervised methods.

What distinguishes anomalies from the rest?

Observe the feature space:

- Higher distance.
- Lower density.
- Easier to separate from the others.

Different methods rely on different criteria.



Unsupervised AD methods

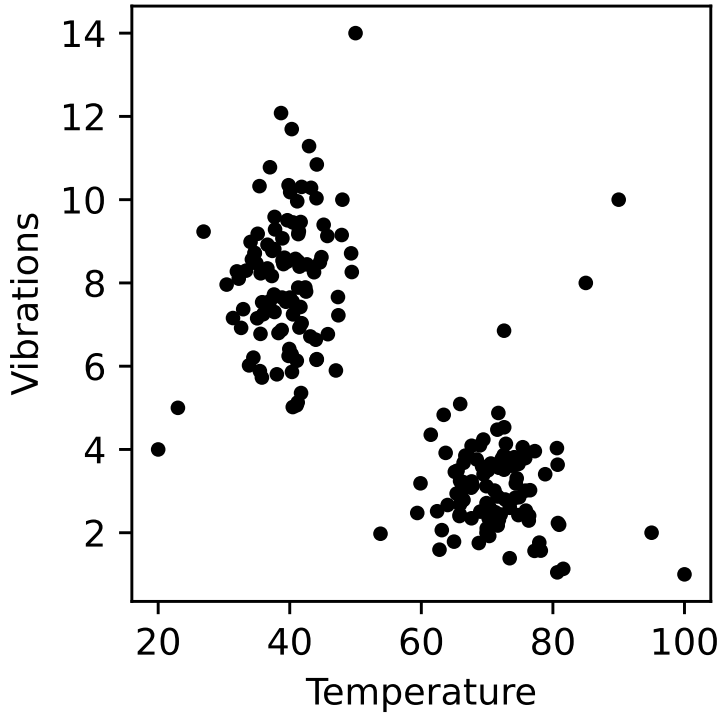
In many real-world systems we have no labels at all ➔ detect anomalies with unsupervised methods.

Week 1:

- Statistical threshold (univariate)
- Linear regression (multivariate)

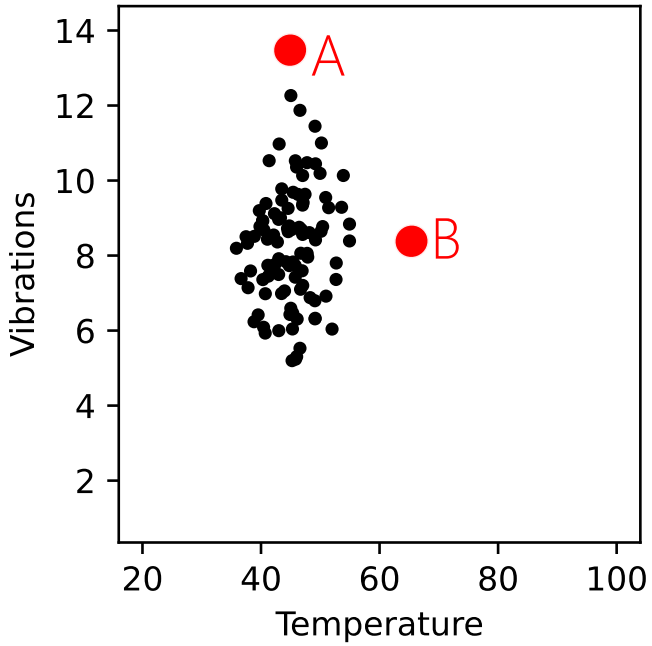
Weeks 3-4:

- Statistical outlier detection (Hotelling T^2)
- Clustering (DBSCAN example)
- One-Class SVM (example)
- Isolation forest (next time)
- Local outlier factor (next time)



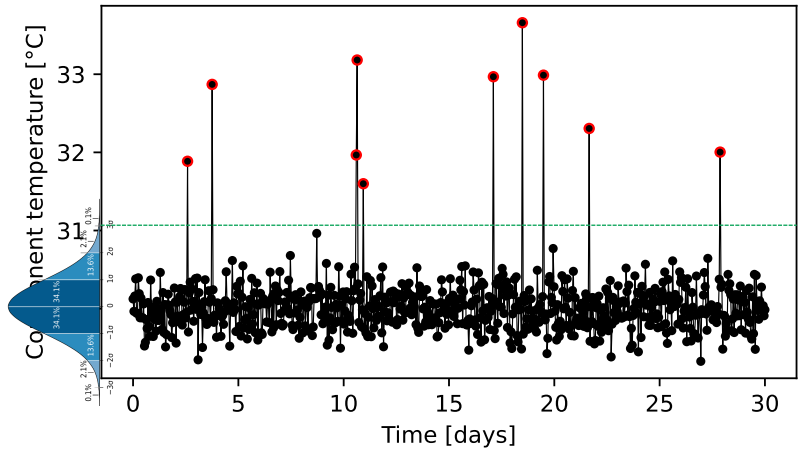
Unsupervised AD: Hotelling's T^2 statistical outlier detector

- Detects outliers by their distance to the normal data.
- Calculate distance while considering the variance and covariance of the normal data.
- First understand how to define distances in a multivariate space.
- Then identify points with a large distance.



Reminder: AD in univariate data

- Example: temperature of an isolated system.
- How to decide on a threshold?
- Use the distribution of “regular” temperatures

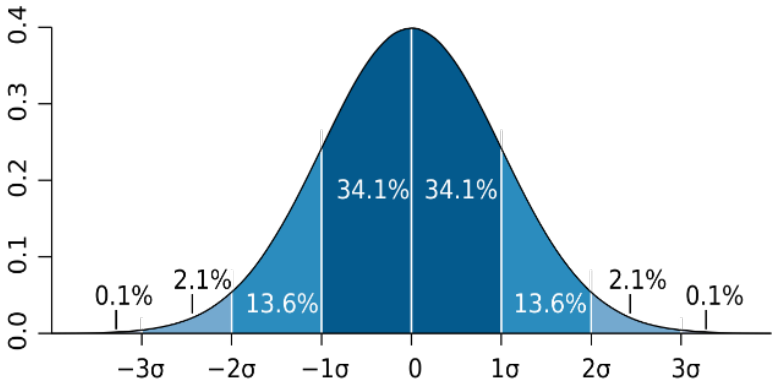


- Model the temperatures with a Gaussian distribution.

➔ 68% of the values are in (μ-σ, μ+σ)

95% are in (μ-2σ, μ+2σ)

99.7% are in (μ-3σ, μ+3σ)



Hotelling's T^2 : z-score for univariate data

- **Univariate data:** measure how unusual a value is by comparing its deviation from the mean, using the natural variability of the data → z-score

$$z = \frac{x - \mu}{\sigma}$$

Intuition:

- if the mean temperature is 20°C and the std is 2°C, then:
  - 21°C ←→ z=0.5 → normal
  - 30°C ←→ z=5 → abnormal
- if the mean temperature is 20°C and the std is 10°C, then:
  - 21°C ←→ z=0.1
  - 30°C ←→ z=1

  → Both are normal

Hotelling's T^2 : From z-score to multiple variables

- **Univariate data:** Z-score measures how unusual a value is by comparing its deviation from the mean, using the natural variability of the data.

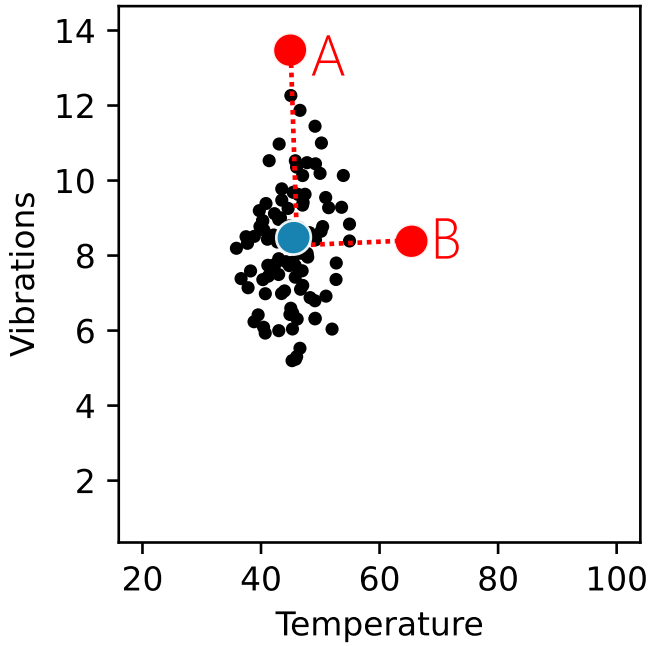
$$z = \frac{x - \mu}{\sigma}$$

Multivariate data?

- Euclidean distance from the mean? Which of A or B is an outlier? $d_A > d_B$

→Variance matters.

→Measure the distance in units of the variance



Hotelling's T^2 : From z-score to multiple variables

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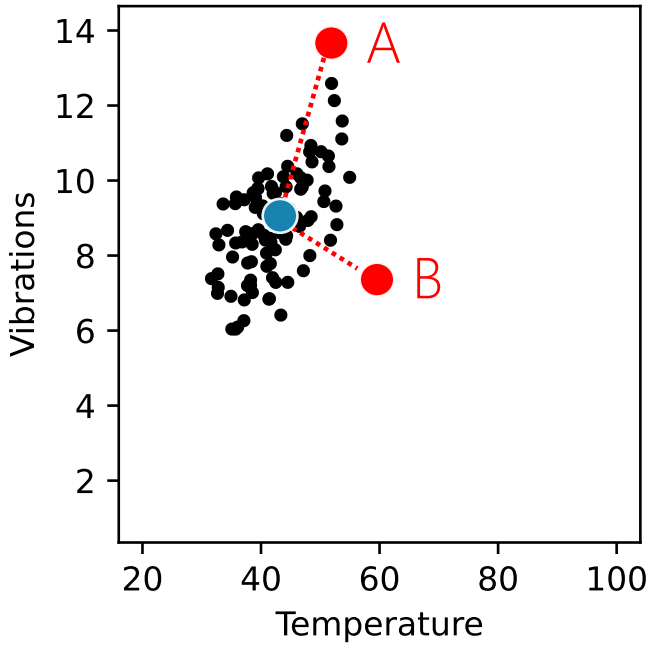
→Variance matters.

→Measure the distance in units of the variance

- What if the variables are correlated?

→Anomalies are samples that violate the correlation.

→Account for covariance when measuring distances



Hotelling's T^2 : The empirical Mahalanobis distance

- **Univariate data:** Z-score measures how unusual a value is by comparing its deviation from the mean, using the natural variability of the data.

$$z = \frac{x - \mu}{\sigma}$$

- Multivariate data: **Mahalanobis distance**

$$d_M^2 = (x - \mu)^T \Sigma^{-1} (x - \mu)$$

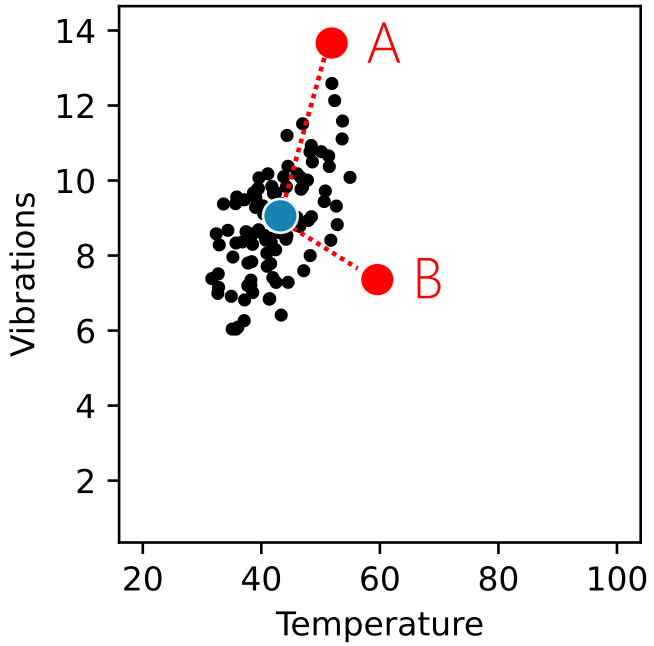
Deviation from the mean

Accounts for variances and correlations

Estimate sample mean and covariance from the data
→ Hotelling's T^2

$$T^2 = (x - \bar{x})^T S^{-1} (x - \bar{x})$$

Small T^2 --> typical observation
Large T^2 --> statistical outlier



How to use Hotelling's T^2

1. Training data → estimate mean and covariance
2. For each new x calculate

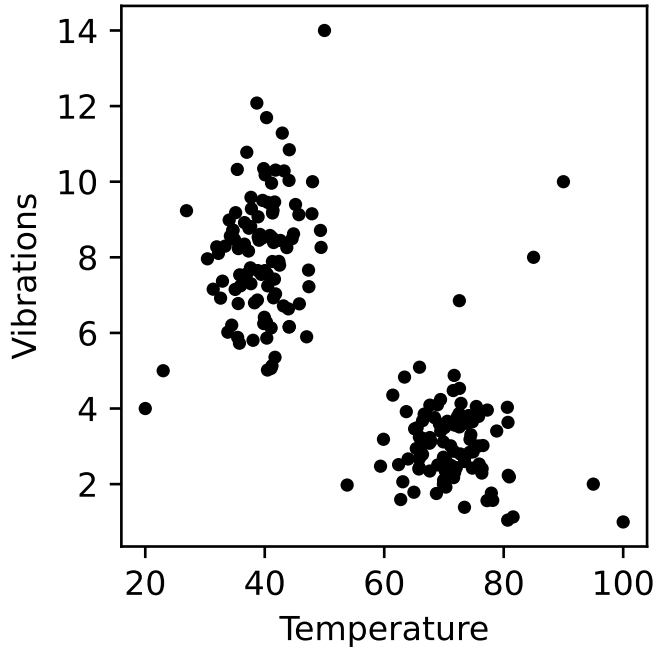
$$T^2 = (x - \bar{x})^T S^{-1} (x - \bar{x})$$

Small T^2 --> typical observation
Large T^2 --> statistical outlier

Pros and Cons of Hotelling's T^2

Pros

- Simple and computationally efficient
- Naturally extends the z-score to multiple variables
- Accounts for variance and correlation
- Produces an interpretable anomaly score



Cons

- Assumes one approximately Gaussian cluster
- Sensitive to outliers in the training data
- Covariance estimation becomes difficult in very high dimensions
- Performs poorly for multimodal normal data (e.g 2 normal regimes).

Clustering with DBSCAN

- Density-Based Spatial Clustering for Applications with Noise.
- Identifies region of high point density in feature space.
- Noise = sparse regions, not meeting the density requirements.
- Intuitive cluster building.
- Anomalies detected as noise.

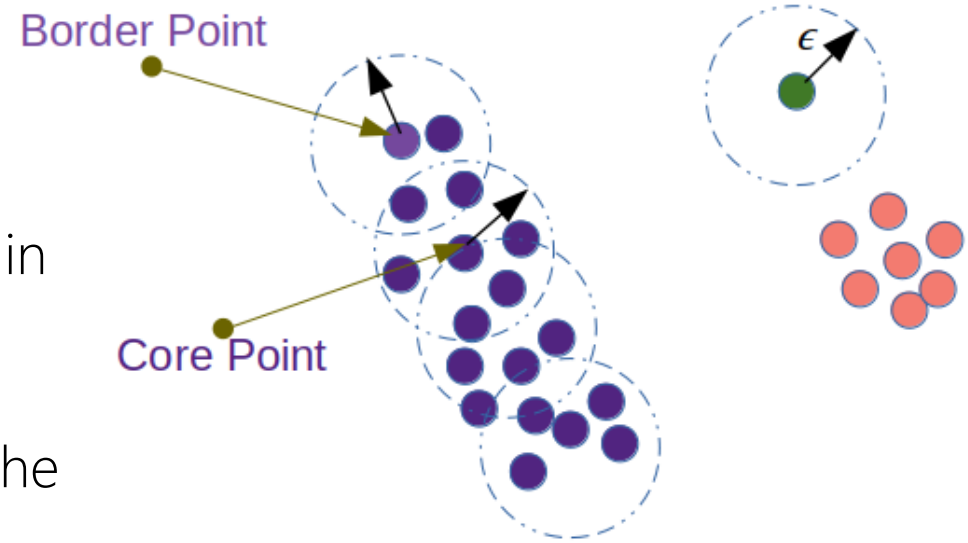
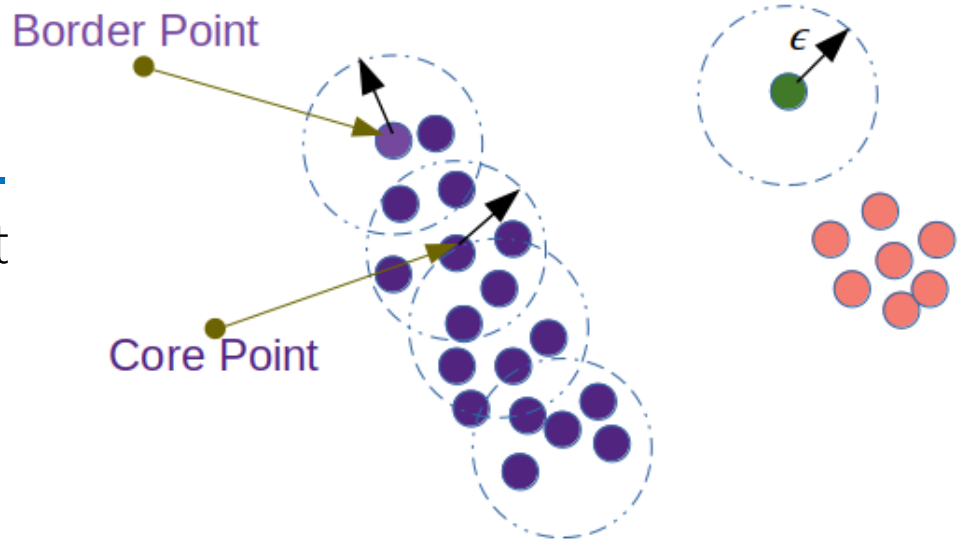


Image source: Towards data science Saptashwa Bhattacharyya.

Clustering with DBSCAN

- Select relevant features.

- Define the neighbourhood radius $\varepsilon = 1$.
- A core point: if its circle includes at least MinPts = 5 neighbours.
- Pick a random core point.
- Extend the cluster by adding all neighbouring core points to it.
- Add all border points (reachable from all core points of this cluster).
- Move to the rest of the points, and build more clusters.
- All points not reachable from any other points are outliers.



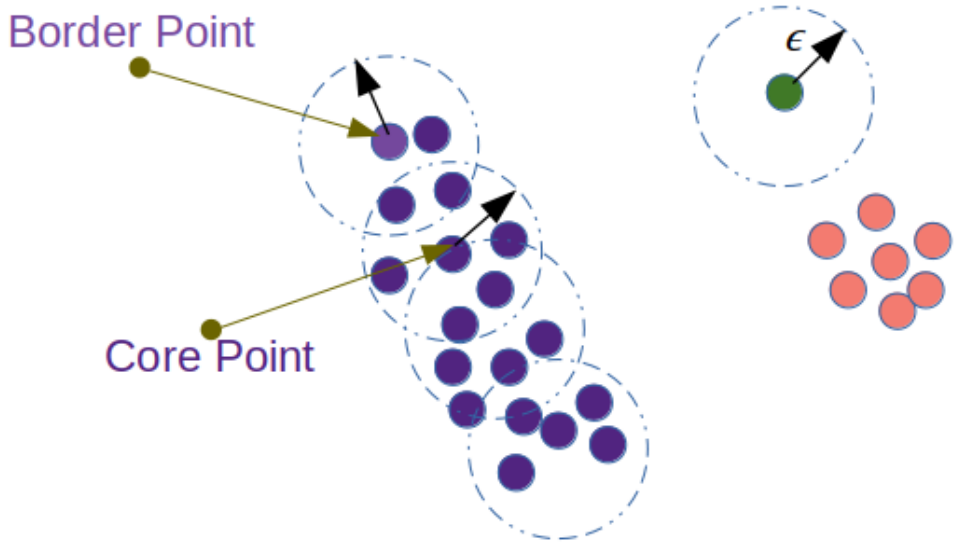
Ester *et al. (1996), A density-based algorithm for discovering clusters in large spatial databases with noise.*

Image source: Towards data science Saptashwa Bhattacharyya.

Hyperparameter estimation

- MinPts has to increase with the data dimensionality. Recommendation: at least 2 x dim.
- Estimate for ε : look for a knee in the distance to the k-NN where k = MinPts.

  (at which distance the typical number of NNs drops suddenly)



Sander et al.(1998). "Density-Based Clustering in Spatial Databases: The Algorithm GDBSCAN and Its Applications".

Why DBSCAN?

- No need to know a priori how many clusters.
- DBSCAN can handle clusters of arbitrary shape, also nested clusters.
- Inherently detects outliers.
- The hyperparameters ε and MinPts can be determined by domain experts that understand the data.

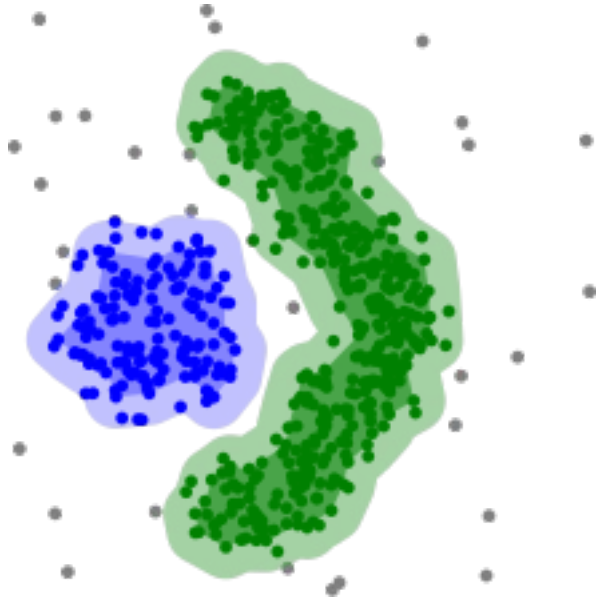
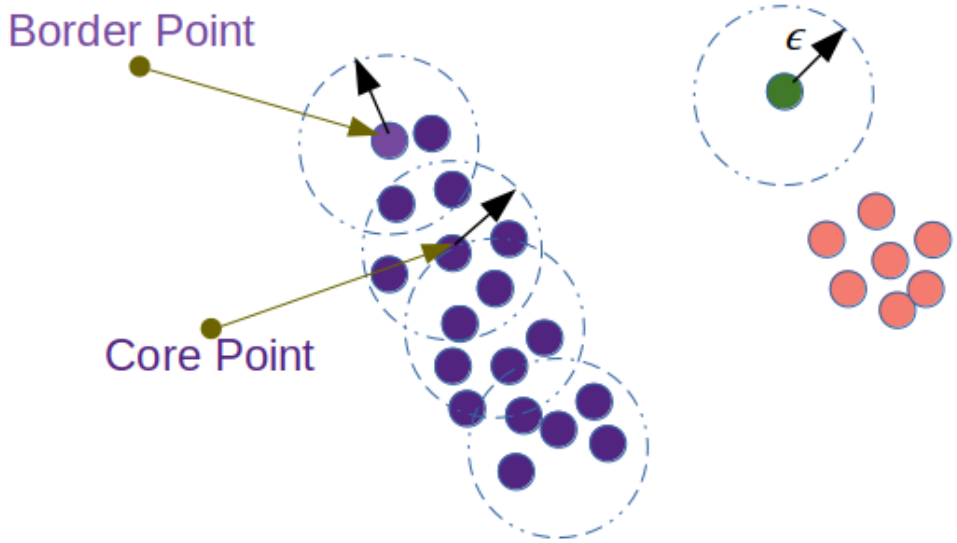


Image source: https://en.wikipedia.org/wiki/DBSCAN

When is DBSCAN not suitable?

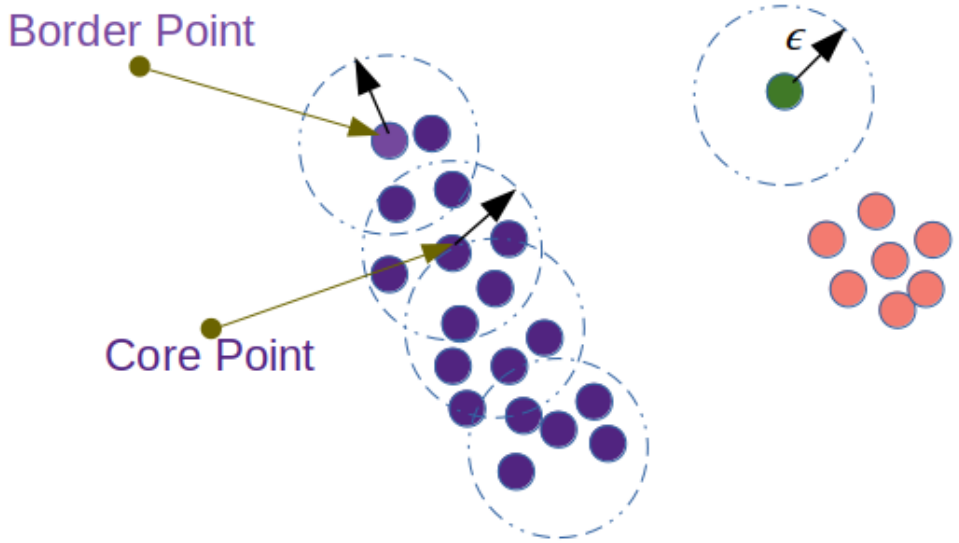
- High dimension ➔ "curse of dimensionality"
- Clusters with very different densities (no single choice of ε and MinPts fits all clusters).
- Without some understanding of the data choosing the hyperparameters can be hard.



Pros and cons of DBSCAN compared to other clustering approaches

- Pro: inherent outlier detection, no need for one abnormal cluster.

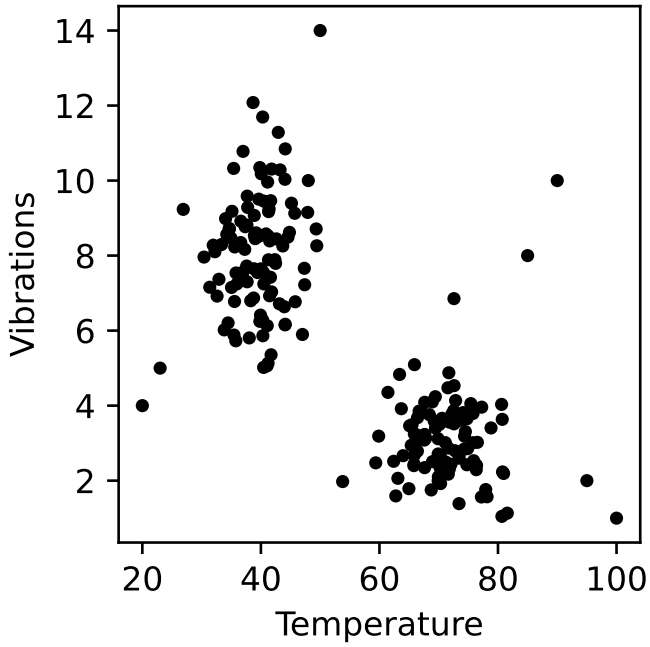
- Con: No inherent predict() rule in the original algorithm.
  - Strict DBSCAN: rerun for each new measurement.
    OR
  - DBSCAN as practical novelty detector: classify a new point as normal if it lies within ε of at least one core point.

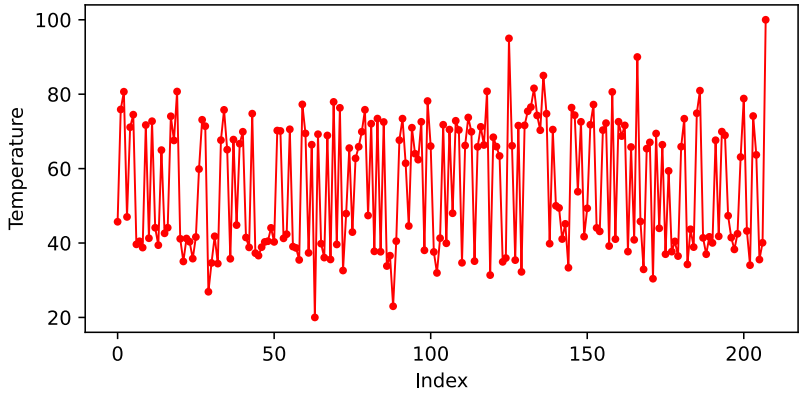


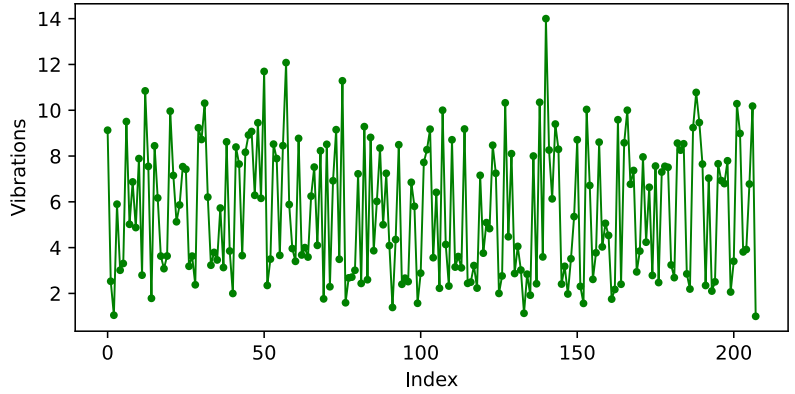
AD with DBSCAN

Monitor the temperature of a component and its vibration amplitude:

- Operating condition 1: low T, intermediate vibrations.
- Operating condition2: High T, low vibrations.







WPI_AnoDet_03a_AD_unsupervised_DBSCAN

Unsupervised AD with One-Class SVM (OC-SVM)

- Instead of deciding by the density, look for a boundary that encapsulates the normal data.

➔Modify the SVM classifer to the case of one class.

➔No need for labels.

Implicit assumptions:

- Few outliers.
- Upper limit on the anomaly ratio.

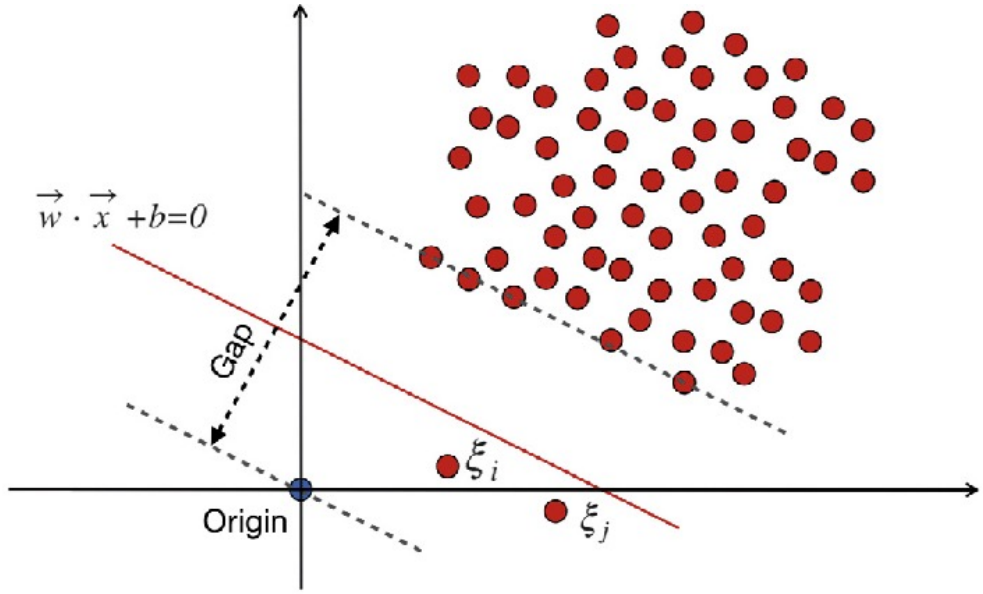


Image source: https://www.analyticsvidhya.com/

Recap SVM classifier

$$\min\left\{\frac{1}{2}\|\vec{w}\|^2 + C\sum_{i=1}^{n}\zeta_i\right\} \quad \text{such that} \quad y_i(\vec{w}\cdot\vec{x}_i + b) \geq 1 - \zeta_i \;\; \forall\vec{x}_i$$

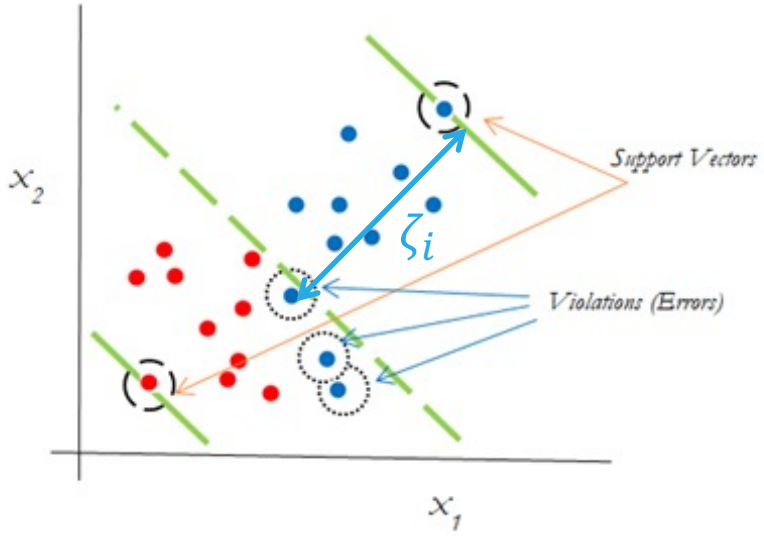
Defines the separating hyperplane $\vec{w}$

Regularization param.

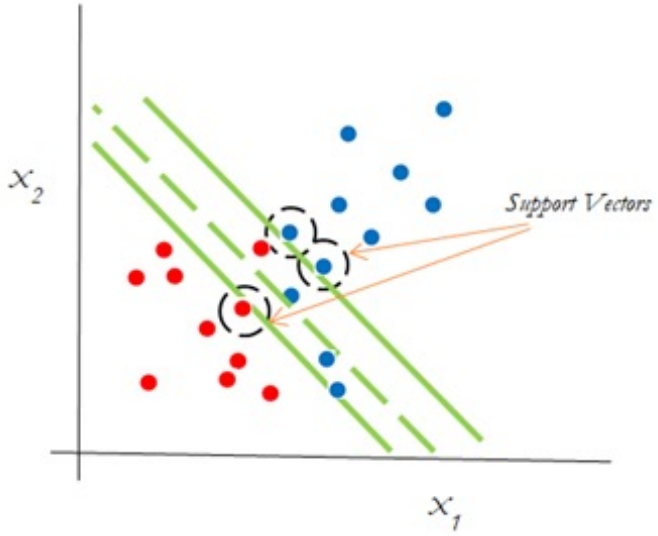
Panelizes misclassifications

$\neq 0$ only for misclassified samples or within margin

Small C: allow many errors



Large C: avoid many errors



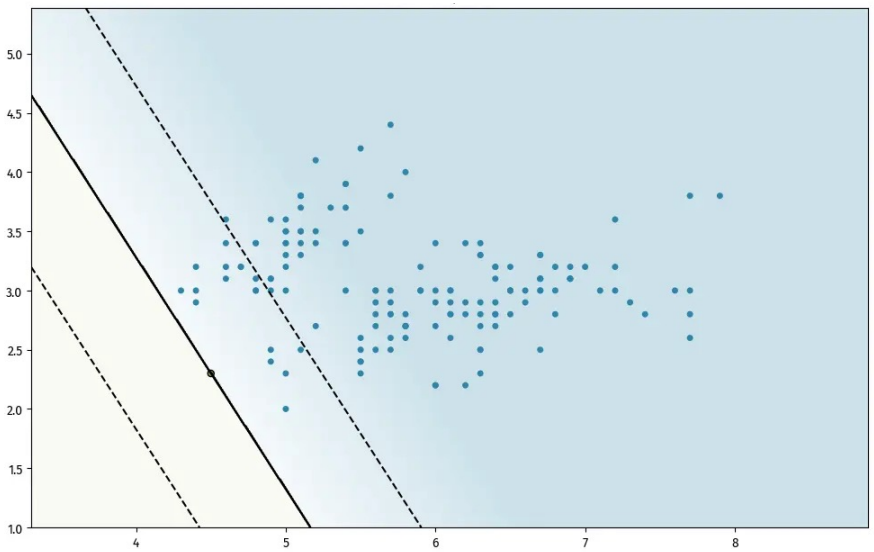
One-Class SVM (OC-SVM)

- SVM maximizes margin between 2 classes.
- Here no labels available ➔ only one class

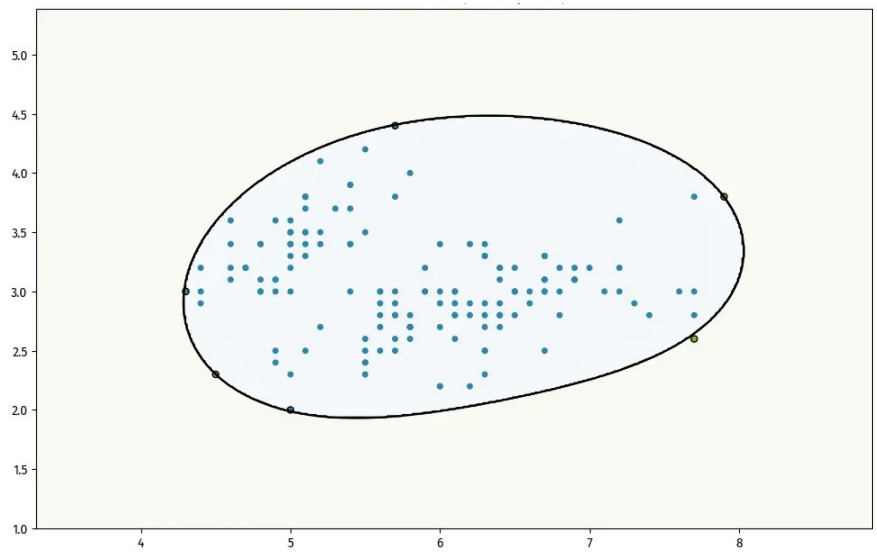
➔ use the 0 point as a second class.

- Or better: in kernel space find a boundary encapsulating all* points.

Linear OCSVM



Radial basis function OCSVM

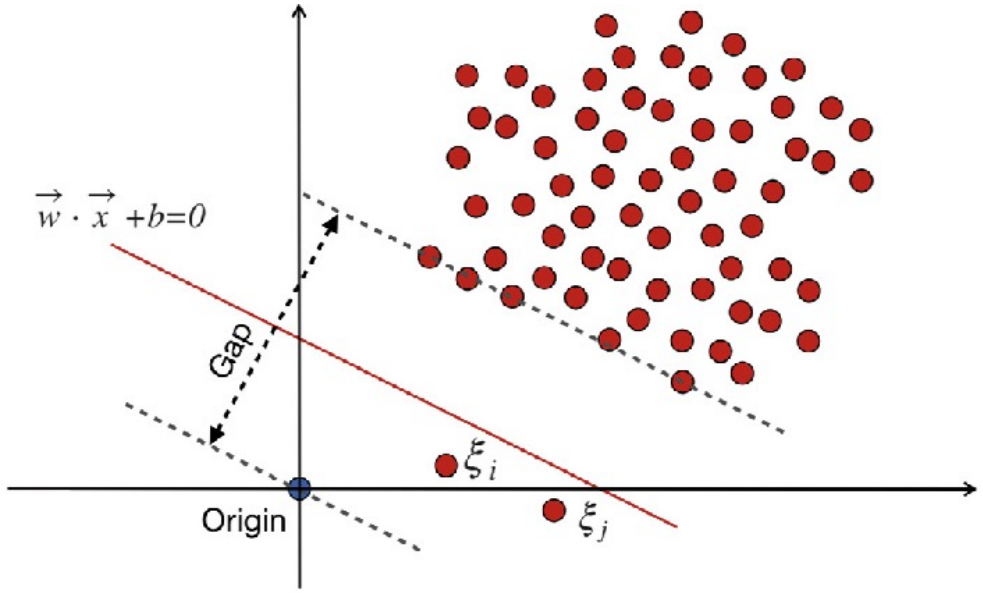


*Allow for slack variables that control the "softness" of the boundary

Image source: https://dm.cs.tu-dortmund.de/en/mlbits/outlier-one-class-svm/

One-Class SVM (OC-SVM)

Find the hyperplane with the maximal margin that separates the data from the origin.



$$\min_{\vec{w},\xi_i,\rho} \left\{ \frac{1}{2}\|\vec{w}\|^2 + \frac{1}{\nu n}\sum_{i=1}^{n}\xi_i - \rho \right\} \text{ such that } \quad y_i(\vec{w}\cdot\vec{x}_i + b) \geq \rho - \xi_i \quad \forall \vec{x}_i$$

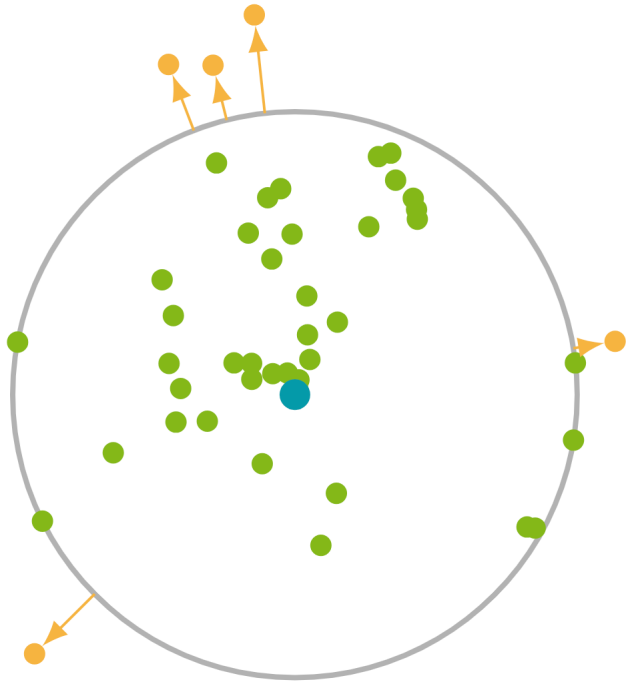
Controls the max. no. of outliers

$\neq \mathbf{0}$ only for outliers or samples within margin

Schölkopf et al. «Support Vector Method for Novelty Detection» (1999)

Support Vector Data Description (SVDD)

- Alternative mathematical formulation of the same idea.
- Point 0 is no longer special.
- Find a sphere that encapsulates all (most) data points.

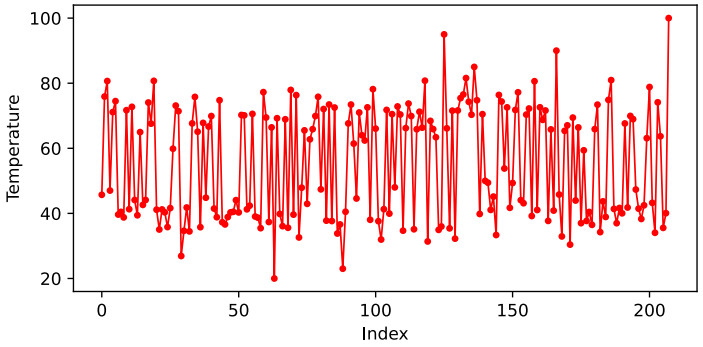


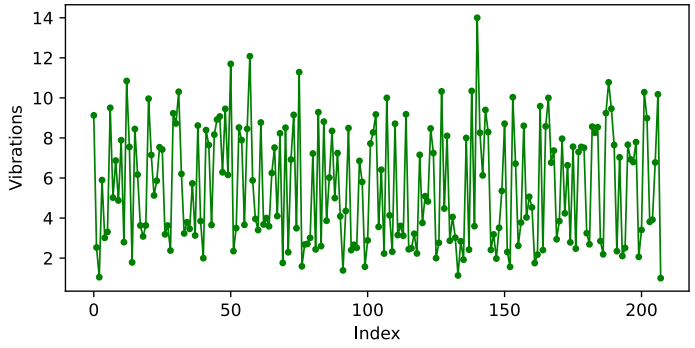
$$\min_{R,\xi_i} \left\{ \frac{1}{2} R^2 + C \sum_{i=1}^{n} \xi_i \right\} \quad \text{such that} \quad \|\vec{x}_i - a\|^2 \leq R^2 + \xi_i \quad \forall \vec{x}_i$$

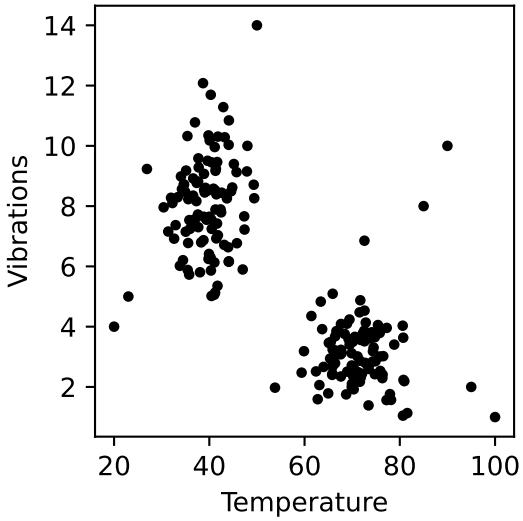
Tax, D.M.J. and Duin, R.P.W. 2004. Support vector data description.

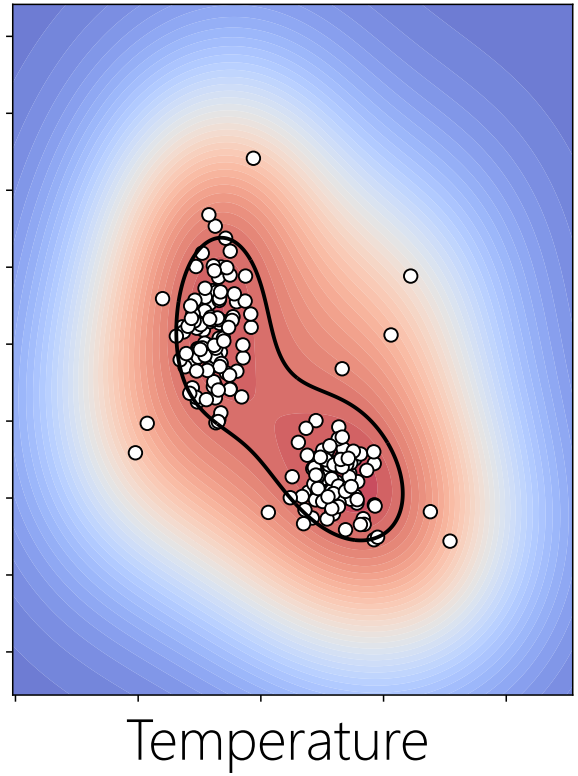
Image source: https://dm.cs.tu-dortmund.de/en/mlbits/outlier-one-class-svm/

Example: AD with OCSVM









WPI_AnoDet_03b_AD_unsupervised_ocsvm

Your exercise: Unsupervised AD in Credit Card Transactions

The datasets contain simulated credit card transactions from one customer. The customer has one or two normal purchasing patterns, depending on the dataset. A small number of transactions are anomalous and do not match the usual behavior.

- Inputs:
  - hour: Hour of the day at which the transaction occurred, between 0 and 24.
  - amount: Transaction amount in CHF.
- Target:
  - Normal or anomaly.
- Task: detect abnormal transactions with
  - DBSCAN
  - OCSVM

RECENT TRANSACTIONS

DATE	AMOUNT	MERCHANT	LOCATION	SCORE	FLAG
May 26, 10:15	$23.50	Coffee House	New York, USA	-0.12	✓
May 26, 12:47	$57.90	SuperMart	New York, USA	-0.15	✓
May 26, 14:05	$89.99	Amazon	New York, USA	-0.18	✓
May 26, 16:20	$1,250.00	ElectroWorld	Miami, USA	-0.72	⚠
May 26, 18:33	$19.90	Pharmacy	New York, USA	-0.10	✓
May 26, 21:17	$2,340.00	Luxury Watch Co.	Paris, France	-0.89	⚠
May 27, 09:02	$25.00	Bakery	New York, USA	-0.11	✓

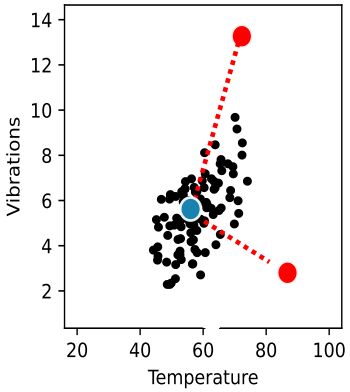
WPI_AnoDet_03c_exercise_AD_unsupervised_DBSCAN_OCSVM

Solution: Unsupervised AD in Credit Card Transactions

- WPI_AnoDet_03d_solution_AD_unsupervised_exercise_DBSCAN_OCSVM

Unsupervised AD methods

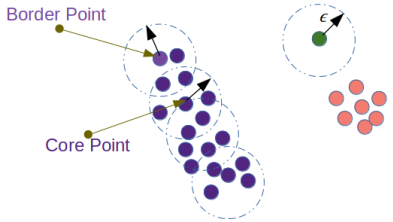
In many real-world systems we have no labels at all → detect anomalies with unsupervised methods.

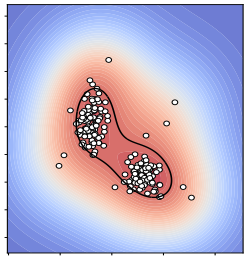


What distinguishes anomalies from the rest?

Observe the feature space:

- Higher distance → Hotelling T^2
- Lower density → DBSCAN
- Easier to separate from the others → OCSVM





Different methods rely on different criteria.

To summarize

- Unsupervised AD: when no labels are available during training.
- Most methods suffer from the “curse of dimensionality”.
- Hyperparameters: heuristics, domain knowledge, some systematic approaches (knee in anomaly scores).
- Evaluation: either with some labeled validation data or apostriori with an expert.
- Note: **Ensemble** methods enhance robustness, and help quantify parameter sensitivity.